

PubMLST and the BIGSdb genomics platform

Keith Jolley



A collection of open-access, curated databases that integrate population sequence data with provenance and phenotype information for over 130 different microbial species and genera.

36,103,520

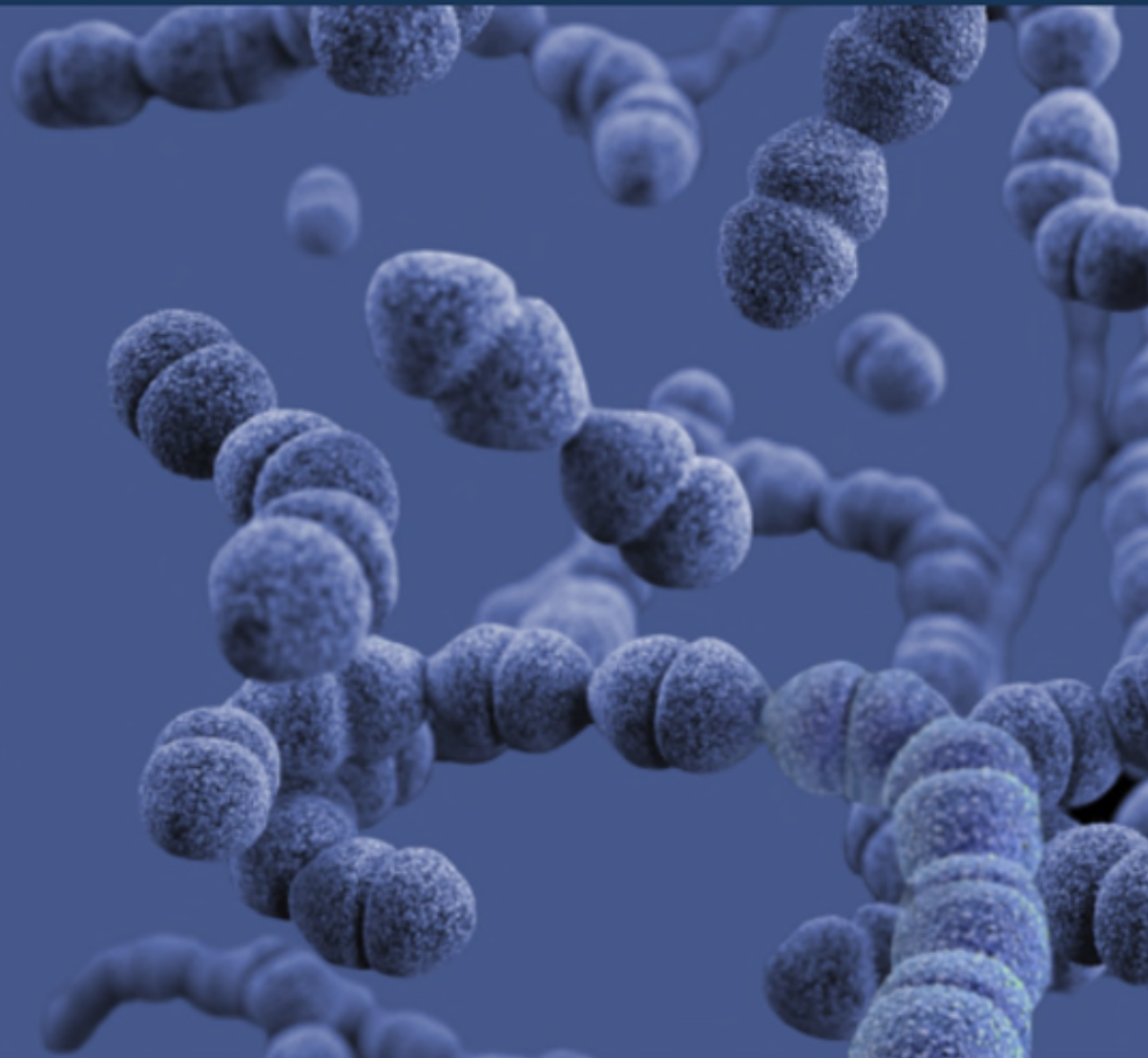
ALLELES

1,275,518

ISOLATES

999,279

GENOMES



PubMLST hosts typing nomenclatures for >130 microorganisms

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Organisms

Most popular



Campylobacter jejuni/coli



Haemophilus influenzae



Neisseria spp.



Staphylococcus aureus



Streptococcus agalactiae



Streptococcus pneumoniae

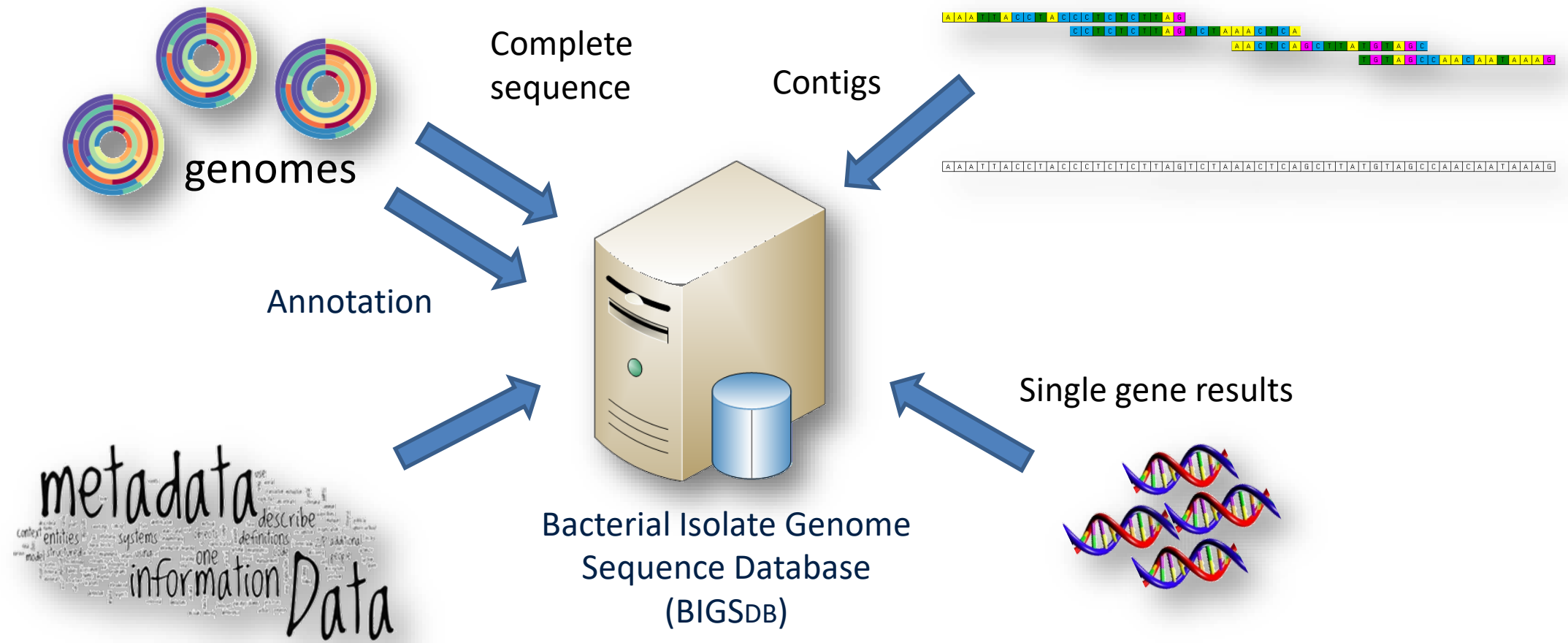
Search

APPLY RESET

A B C D E F G H K L M N O P R S T U V W X Y

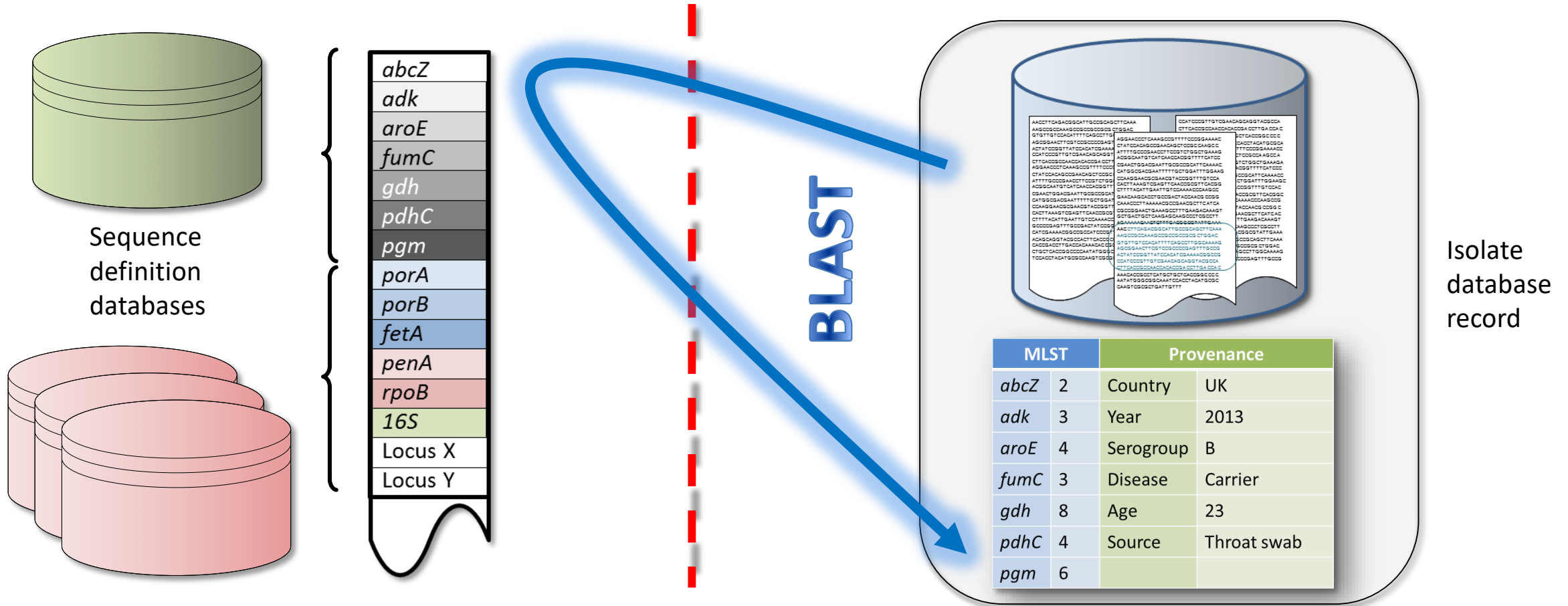
136 organisms

Population genomics: the BIGSdb platform



Jolley & Maiden 2010. BIGSdb: Scalable analysis of bacterial genome variation at the population level. *BMC Bioinformatics* **11**:595

Bacterial Isolate Genome Sequence Database (BIGSdb) design philosophy



Jolley & Maiden 2010. BIGSdb: Scalable analysis of bacterial genome variation at the population level. *BMC Bioinformatics* **11**:595

BIGSdb consists of two main database structures

Sequence definitions



Allele1: TTTGATACTGTTGCCGAAGGTTTCCC
Allele2: TTTGATACCGTTGCCGAAGGTTTCCC
Allele3: TTTGATTCCGTTGCCGAAGGTTTCCC
Allele4: TTTGATTCCGATGCCGAAGGTTTCCC

Allelic profiles

ST	abcZ	adk	aroE	fumC	gdh	pdhC	pgm	clonal_complex
1	1	3	1	1	1	1	3	ST-1 complex
2	1	3	4	7	1	1	3	ST-1 complex
3	1	3	1	1	1	23	13	ST-1 complex
4	1	3	3	1	4	2	3	ST-4 complex
5	1	1	2	1	3	2	3	ST-5 complex
6	1	1	2	1	3	2	11	ST-5 complex
7	1	1	2	1	3	2	19	ST-5 complex
8	2	3	7	2	8	5	2	ST-8 complex
9	2	3	8	10	8	5	2	ST-8 complex
10	2	3	4	2	8	15	2	ST-8 complex
11	2	3	4	3	8	4	6	ST-11 complex
12	4	3	2	16	8	11	20	

Nomenclature

BIGSdb consists of two main database structures

Provenance



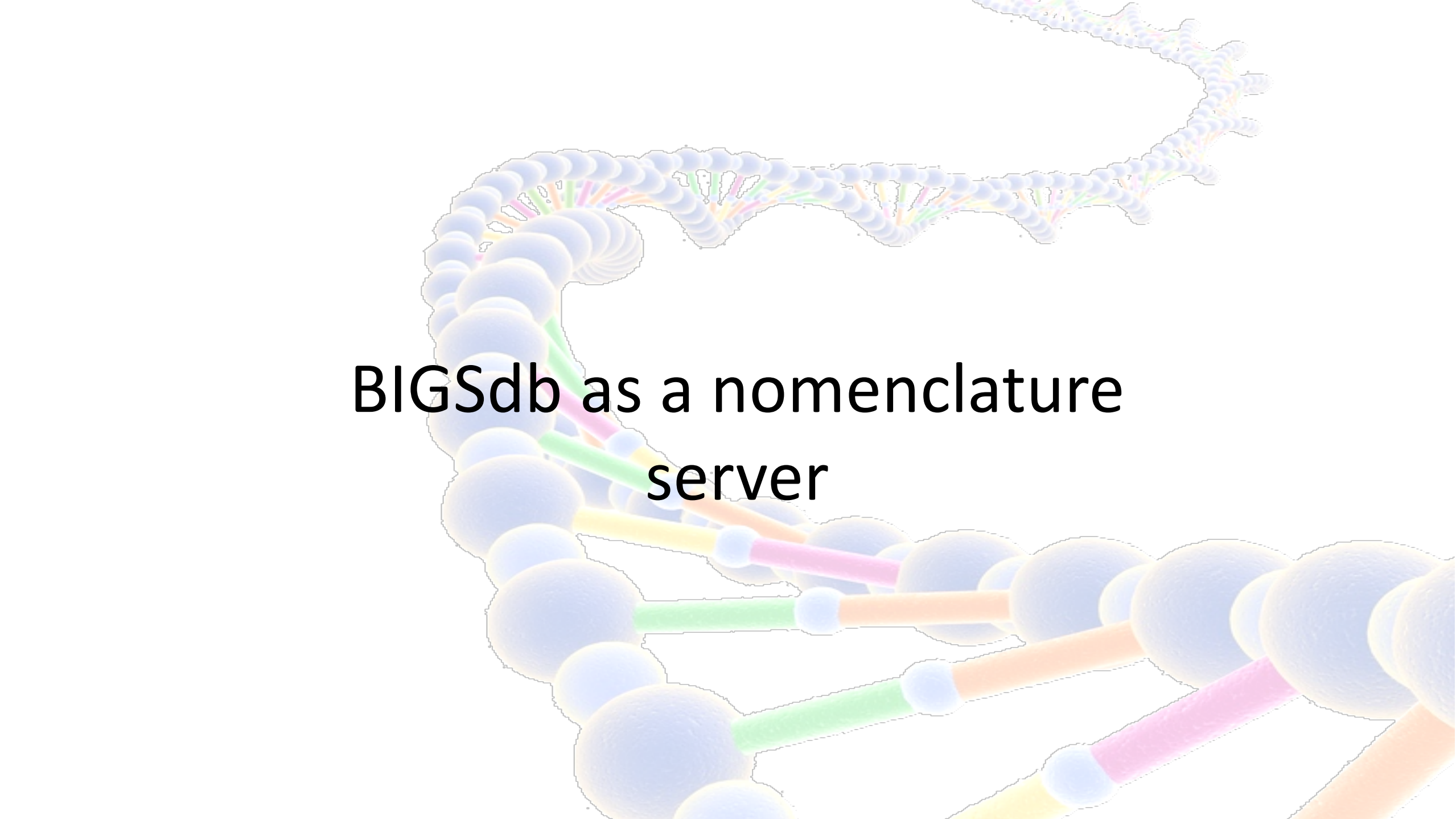
Phenotype



Sequences



Isolate data

A 3D molecular model of a DNA double helix is shown, rendered with semi-transparent blue spheres for the sugar-phosphate backbone and colored sticks (green, orange, pink, yellow) for the nitrogenous base pairs. The DNA strand is curved, and the text is centered over it.

BIGSdb as a nomenclature
server

BIGSdb is a nomenclature server

PubMLST

Public databases for molecular typing
and microbial genome diversity

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Home > Organisms > *Neisseria* spp. > *Neisseria* typing > Sequence query



Sequence query

Please paste in your sequence to query against the database. Query sequences will be checked first for an exact match against the chosen (or all) loci - they do not need to be trimmed. The nearest partial matches will be identified if an exact match is not found. You can query using either DNA or peptide sequences. 

Please select locus/scheme

Order results by

MLST

locus

Enter query sequence (single or multiple contigs up to whole genome in size)

Alternatively upload FASTA file

or enter Genbank accession

```
>187372 NODE_1118_length_222_cov_28.878378
TCCCTGTGGGAGAGGGCTAGGGAGAGGGCGGCCAAACCGCAGGTTTGCTTGGGCGGCATTT
TCAACGTGCAGGCTGCTTTTCGATTTTTGCAGCTTCGGTTTTAGCTTCGCAGAAACTCT
GCTTCCTTCGAAAGCTCCGTTTTTCAGACGACCTTTCAGTTTTTCAGTGCACACAGGCTTG
TGAAAGCGTTATCGGCTTTGATGTAAGCCTACGGCTTACTTTCCCCCCCCCCCCCCCC
CCCCCCCCCCCCCCCCCGGG
>187373 NODE_320_length_7232_cov_48.889935
```

Select FASTA file:

Browse...

No file selected.

Action

RESET

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BIGSdb is a nomenclature server

7 exact matches found.

Locus	Allele	Length	Contig	Start position	End position	Flags	Comments
abcZ	2	433	187414	5637	6069		
adk	3	465	187432	8935	9399		
aroE	4	490	187449	4826	5315		
fumC	3	465	187396	4871	5335		
gdh	8	501	187466	7333	7833		
pdhC	4	480	187542	56831	57310		
pgm	6	450	187395	21997	22446		

Only exact matches are shown above. If a locus does not have an exact match, try querying specifically against that locus to find the closest match.



MLST



Matching profile

ST: 11

clonal complex: ST-11 complex

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Closest matching genomes can be identified by cgMLST clustering

PubMLST

Public databases for molecular typing and microbial genome diversity

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Home > Organisms > *Neisseria* spp. > *Neisseria* typing > Sequence query

Help

Sequence query

Please paste in your sequence to query against the database. Query sequences will be checked first for an exact match against the chosen (or all) loci - they do not need to be trimmed. The nearest partial matches will be identified if an exact match is not found. You can query using either DNA or peptide sequences. ⓘ

Please select locus/scheme

Order results by

N. meningitidis cgMLST v1.0

locus

Enter query sequence (single or multiple contigs up to whole genome in size)

Alternatively upload FASTA file

or enter Genbank accession

GATTTTCTCGGTTTTTCCAGCTTATCGACCAAGTCTTGCGAGGAATACGCGCGGATAAT
GCCCGTCCCTTTTGCCGCGCGGTTTAAAAATTCGTCCACCTTACCGATTGTTGCGTTGTA
GGCGACAACCTTAAATCCGCAATCGTTCATATTCAAAATCAGGTTTTGCCCCATAACCGC
CAAACCGATTACGCCAATATCGCCTTTCATTGCAGGAAGCTCCGTTATAGATTTAATTTA
TCGACCGCAACTCTACCTGATTACACTTGTTTAAACAATCCTTAACCTTTTAATTTTGG
AAAAGATGCCTTTACGCTTTACCGTGCGTTTCCCTGAAGGC

Select FASTA file:

Browse...

No file selected.

Action

RESET

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Closest matching genomes can be identified by cgMLST clustering

N. meningitidis cgMLST v1.0



Matching profiles

Closest profile: [cgST-56](#)

Mismatches: 5

Loci matched: 1600/1605 (99.7%)



Similar profiles (determined by classification schemes)

Experimental schemes are subject to change and are not a stable part of the nomenclature.

Classification scheme	Clustering method	Mismatch threshold	Status	Group	Profiles	Isolates
Nm_cgc_200 ⓘ	Single-linkage	200	experimental	65	4810	PubMLST isolates 3474
Nm_cgc_100 ⓘ	Single-linkage	100	experimental	42	1	PubMLST isolates 1
Nm_cgc_50 ⓘ	Single-linkage	50	experimental	50	1	PubMLST isolates 1
Nm_cgc_25 ⓘ	Single-linkage	25	experimental	50	1	PubMLST isolates 1

Contact

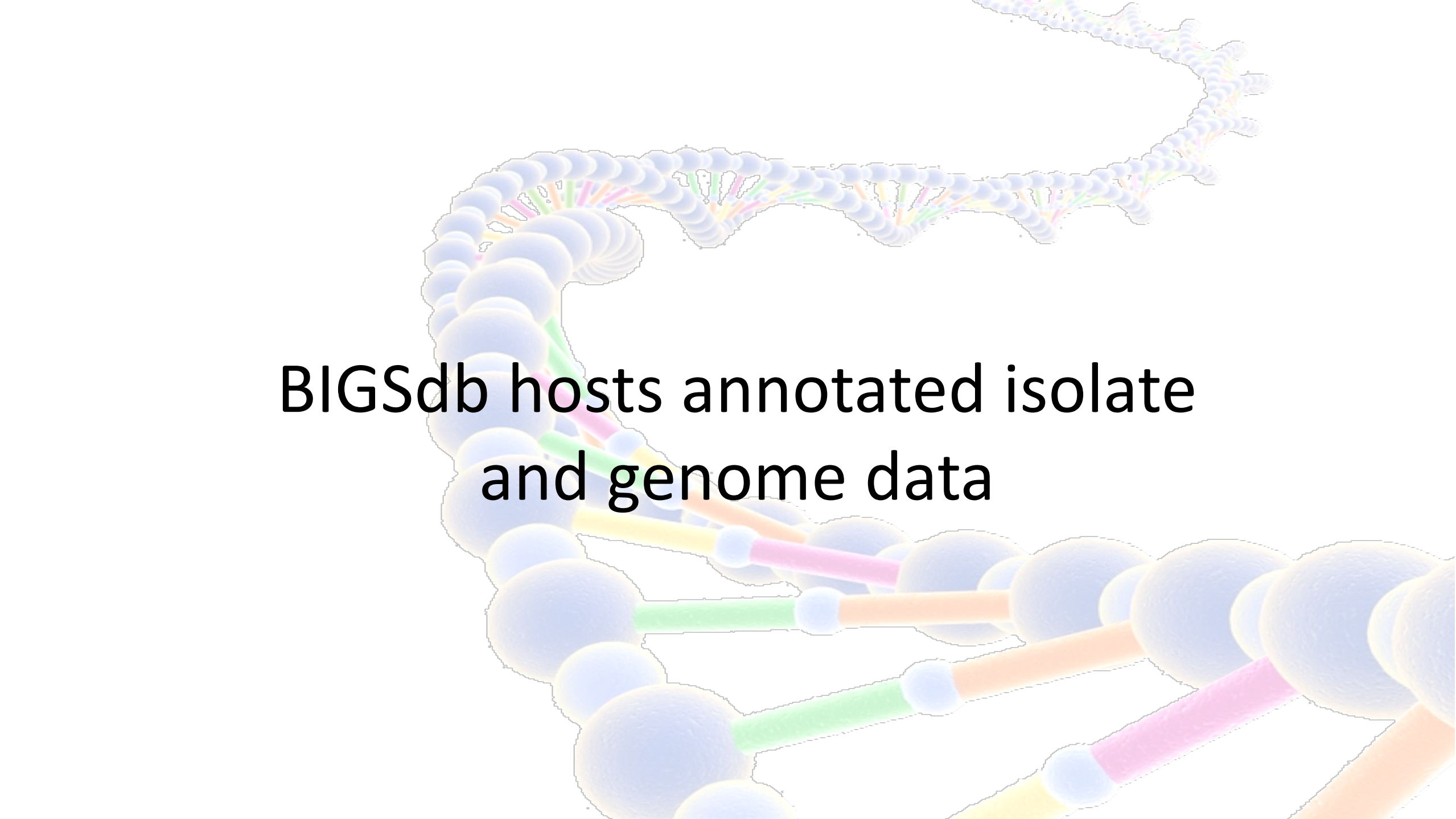
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BIGSdb hosts annotated isolate
and genome data

Annotation links genomes with provenance and phenotype

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Home > Organisms > *Neisseria* spp. > *Neisseria* isolates > Isolate information

 [Help](#) 

Full information on isolate 7891 (id:7)



Projects

This isolate is a member of the following project:

107 global collection

This dataset was originally used to validate MLST and was chosen to represent global diversity of *N. meningitidis* in the latter half of the Twentieth Century. It has been used in many publications since and the isolates are available as the EMGM MLST reference collection.



Provenance/primary metadata

id: 7	genogroup: A	sender: Wendell Zollinger, Dept Bacterial Diseases, Walter Reed Army Institute of Research, Washington DC, USA
isolate: 7891	genogroup notes: A backbone: All essential capsule genes intact and present. Prediction code: https://github.com/ntopaz/characterize_neisseria_capsule .	curator: Auto Tagger
aliases: B54; NIBSC_2760; Z1054	capsule group: A	update history: 122 updates show details
strain designation: A: P1.20,9: F3-1: ST-5 (cc5)	MLEE designation: Subgroup III	date entered: 2001-02-07
country: Finland	serotype: 4,21	datestamp: 2020-10-06
continent: Europe	sero subtype: P1.9	
year: 1975	ET no: 48	
disease: invasive (unspecified/other)	biosample accession: ERS006946 www.ebi.ac.uk 	
source: CSF	comments: Pili I,IIa	
epidemiology: epidemic		
species: <i>Neisseria meningitidis</i>		
serogroup: A		

Projects

Provenance

Annotation links genomes with provenance and phenotype

Phenotype



Secondary metadata



Vaccines

Bexsero reactivity: exact match

notes

Bexsero notes: NadA_peptide: 8 is exact match to vaccine variant - peptide sequence match (PMID:27521232)

Trumenba reactivity: insufficient data

notes



Publications (8)

- Bennett JS, Jolley KA, Sparling PF, Saunders NJ, Hart CA, Feavers IM, Maiden MC (2007). Species status of *Neisseria gonorrhoeae*: evolutionary and epidemiological inferences from multilocus sequence typing. *BMC Biol* 5:35 **576 isolates**
- Bratcher HB, Corton C, Jolley KA, Parkhill J, Maiden MC (2014). A gene-by-gene population genomics platform: de novo assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes. *BMC Genomics* 15:1138 **108 isolates**
- Didelot X, Urwin R, Maiden MC, Falush D (2009). Genealogical typing of *Neisseria meningitidis*. *Microbiology* 155:3176-86 **93 isolates**



Sequence bin

contigs: 199
total length: 2,057,385 bp
max length: 112,831 bp

mean length: 10,339 bp
N50 contig number: 29
N50 length (L50): 23,361

N90 contig number: 94
N90 length (L90): 6,218
N95 contig number: 116

N95 length (L95): 3,616
loci tagged: 2,213

Show sequence bin

Publications

Genome
sequence

Annotation links genomes with provenance and phenotype

Clusters



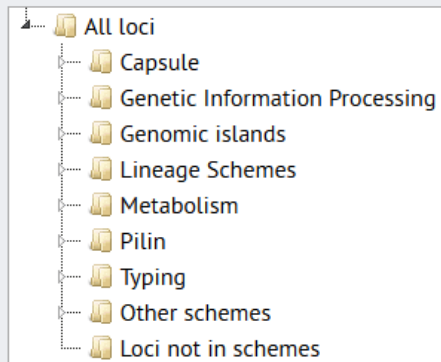
Similar isolates (determined by classification schemes)

Experimental schemes are subject to change and are not a stable part of the nomenclature.

Classification scheme	Underlying scheme	Clustering method	Mismatch threshold	Status	Group
Nm_cgc_200 ⓘ	N. meningitidis cgMLST v1.0	Single-linkage	200	experimental	group: 1 (360 isolates)
Nm_cgc_100 ⓘ	N. meningitidis cgMLST v1.0	Single-linkage	100	experimental	group: 5 (5 isolates)



Schemes and loci



Navigate and select schemes within tree to display allele designations



Tools



Analysis: **rMLST species id** **PCR**

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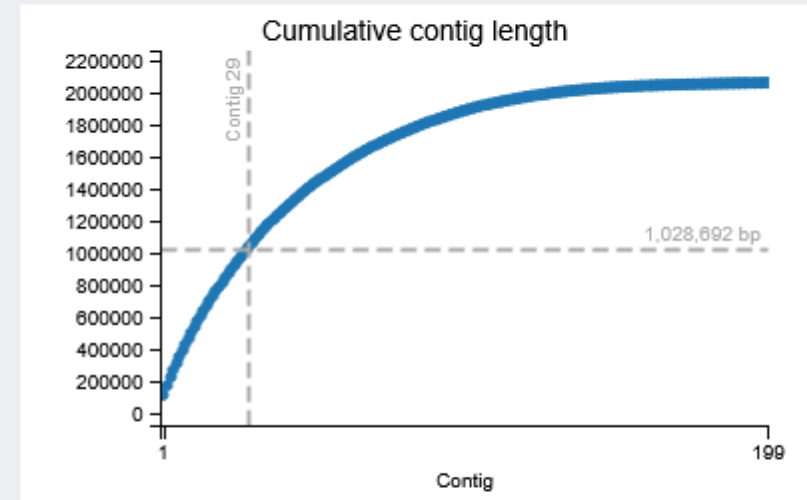
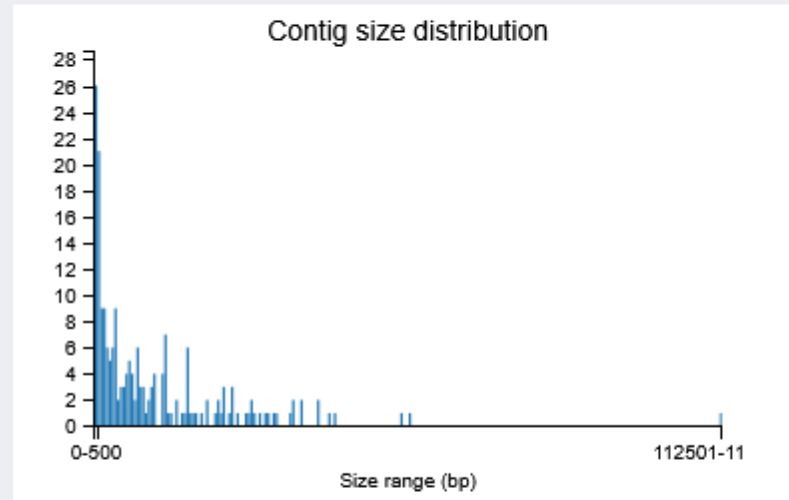


Assembly metrics are available for records with genomes

Sequence bin for 7891

Contig summary statistics

Contigs: 199
Total length: 2,057,385
Minimum length: 267
Maximum length: 112,831
Mean length: 10,339
N50: 23,361
L50: 29
N90: 6,218
L90: 94
N95: 3,616
L95: 116



Completeness of schemes containing only core loci can be used as an assembly quality metric



Annotation quality metrics

Provenance information

Fields used in metric		Fields completed	Annotation	
			Score	Status
5		4	80	—

Missing field values for: age_range

Scheme completion

Scheme	Scheme loci	Designated loci	Annotation	
			Score	Status
rplF species	1	1	100	✓
Finotyping antigens	3	3	100	✓
Bexsero Antigen Sequence Typing (BAST)	5	5	100	✓
MLST	7	7	100	✓
Ribosomal MLST	53	53	100	✓
N. meningitidis cgMLST v3	1329	1318	99	✓
N. meningitidis cgMLST v2	1422	1411	99	✓

Datasets can be searched by wide-range of criteria

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Home > Organisms > *Neisseria* spp. > *Neisseria* isolates > Search or browse database

Search or browse database

Enter search criteria or leave blank to browse all records. Modify form parameters to filter or enter a list of values.

Isolate provenance fields

Combine with: **AND**

country = France

year >= 2010

Filters

Publication: Select options

Project: Select options

MLST profiles: complete

Clonal complex (MLST): ST-41/44 complex

Sequence bin: Sequence bin size >= 2 Mbp

☐ Include old record versions

Add filter:

Add

Display/sort options

Order by: id

ascending

Display: 25 records per page

Action

RESET

SEARCH

83 records returned (1 - 25 displayed). Click the hyperlinks for detailed information.

Your projects

Select project...

Add these records

Bookmark query

2020-10-19:1

Add bookmark

<<

<

1

2

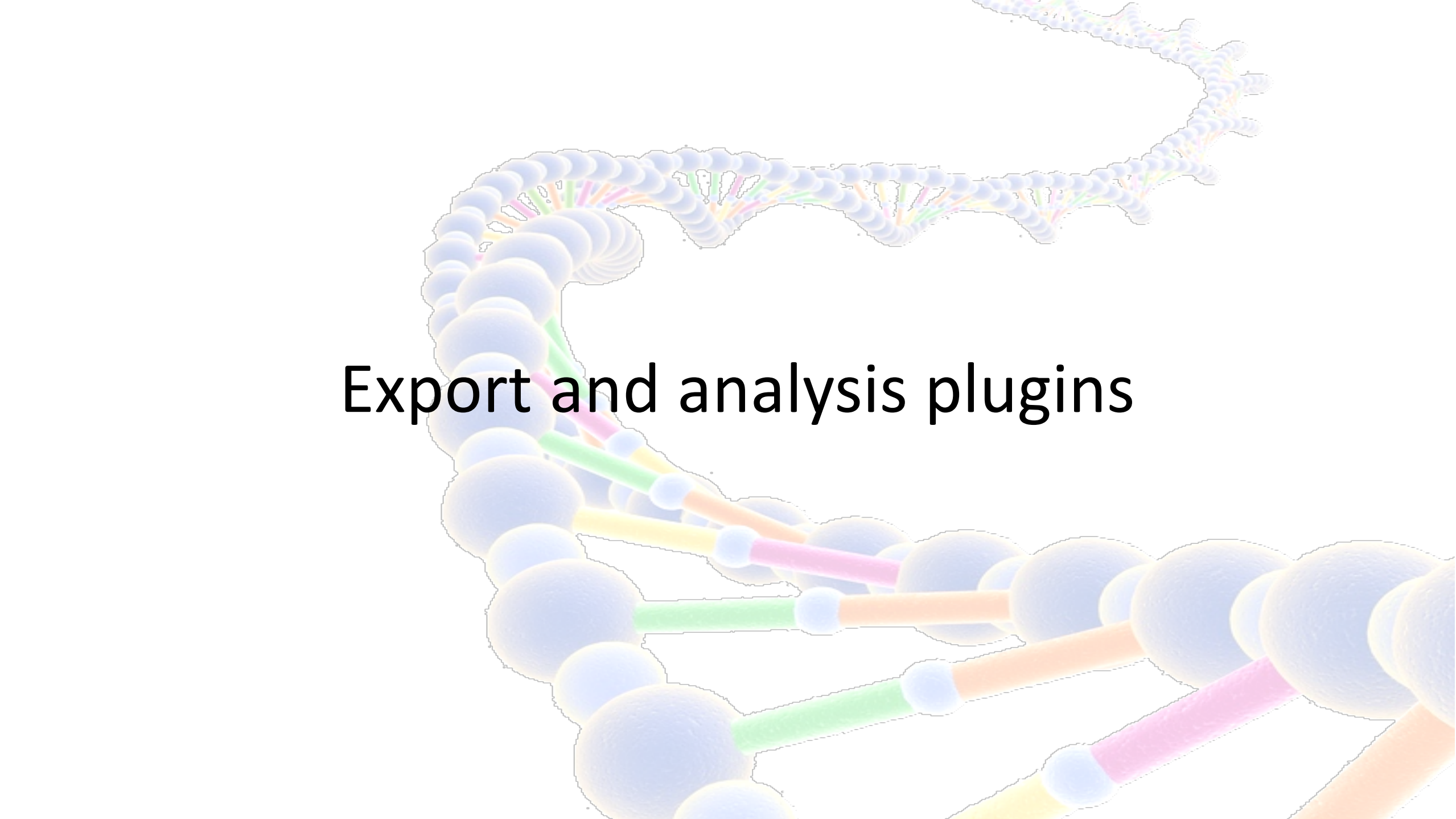
3

4

>

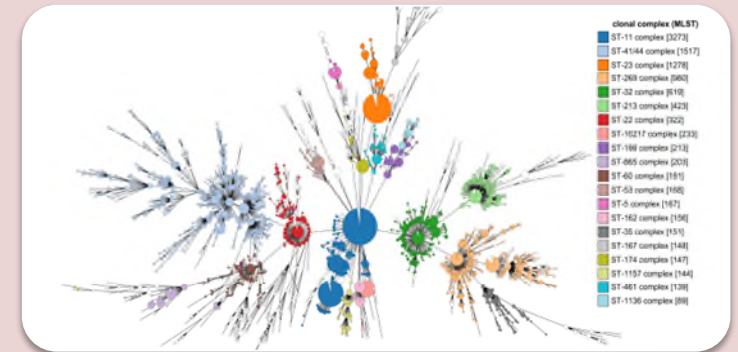
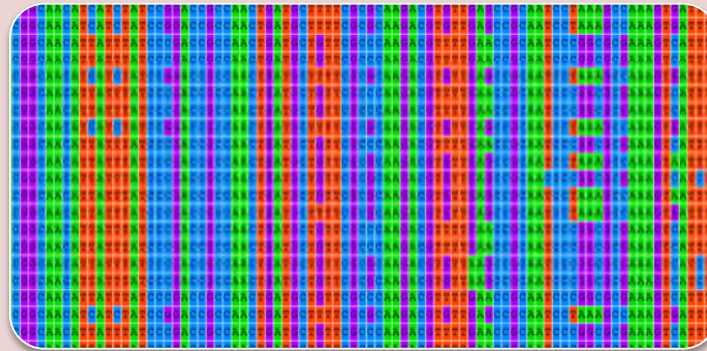
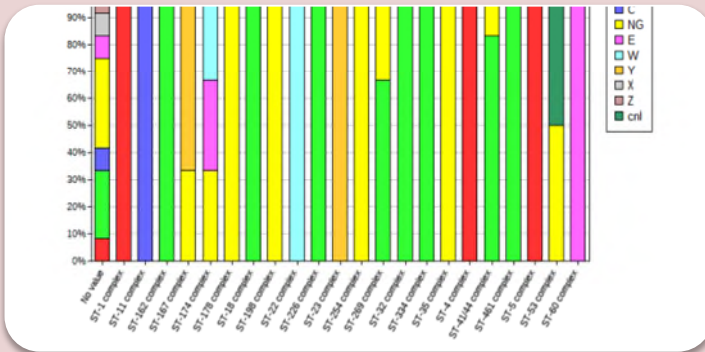
>>

Isolate fields														Seqbin size (bp)	MLST		Finotyping antigens							
id	isolate	aliases	country	region	year	date sampled	date received	non culture	age yr	age range	age mth	sex	disease	source	species	capsule group	Bexsero reactivity	Trumenba reactivity	ST	clonal complex	PorA_VR1	PorA_VR2	FetA_VR	
39963	LNP27359		France		2013										Neisseria meningitidis	B	exact match	insufficient data	2,139,700	1403	ST-41/44 complex	7-2	4	F1-5
39966	LNP27364		France		2013										Neisseria meningitidis	B	insufficient data	cross-reactive	2,102,024	3532	ST-41/44 complex	7-1	1	F1-88
39970	LNP27374		France		2013										Neisseria meningitidis	B	exact match	cross-reactive	2,069,422	1403	ST-41/44 complex	7-2	4	F1-5
39976	LNP27388		France		2013										Neisseria meningitidis	B	exact match	cross-reactive	2,075,331	41	ST-41/44 complex	7-2	4	F1-5
39993	LNP27432		France		2013										Neisseria meningitidis	B	exact match	cross-reactive	2,112,922	41	ST-41/44 complex	7-2	4	F1-5
40009	LNP27478		France		2013										Neisseria meningitidis	B	exact match	cross-reactive	2,071,915	7385	ST-41/44 complex	7-2	4	F1-5
40011	LNP27486		France		2014										Neisseria	B	insufficient	insufficient	2,136,451	10685	ST-41/44	7-1	15-1	F3-6



Export and analysis plugins

Plugins



Breakdown

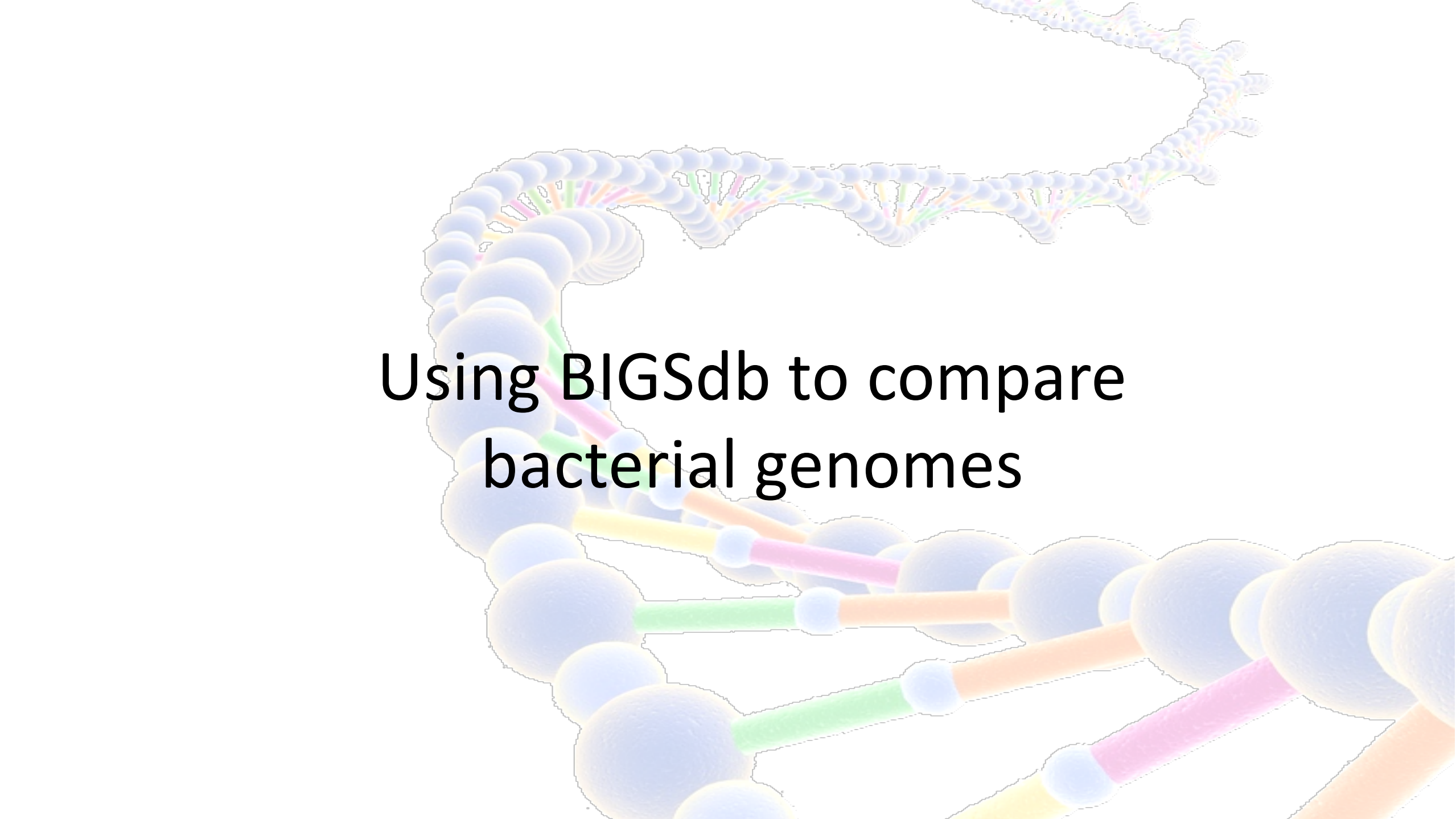
- Single fields
- Pivot tables (2 fields)
- Combinations
- Polymorphic sites
- Sequence bin

Export

- Isolate dataset
- Contigs
- Aligned sequences

Analysis

- Genome Comparator
- GrapeTree
- PhyloViz
- iTOL
- Microreact
- In silico PCR



Using BIGSdb to compare
bacterial genomes

Genomes can be rapidly compared using gene-by-gene variant numbering approach

Locus	Product	Sequence length	Genome position	Reference genome	644 (L93/4286)	662 (2837)	665 (2845)	666 (2843)	667 (2842)	669 (2846)	670 (2840)	671 (2844)	672 (2847)	698 (FAM18)	41784 (2839)	41785 (2838)
NMC0001 lpxC envA	UDP-3-O-[3-hydroxymyristoyl] N-acetylglucosamine deacetylase	924	1261	1	1	1	1	1	1	1	1	1	1	1	1	2
NMC0002 pilS1	pilin (fragment)	291	3341	1	1	1	1	1	1	1	1	1	1	1	2	1
NMC0003 pilS2	truncated pilin	366	3675	1	2	2	2	2	2	2	2	2	2	1	2	2
NMC0004 fbp	peptidyl-prolyl cis-trans isomerase	330	4069	1	2	2	2	2	2	1	2	2	2	1	2	2
NMC0005	putative membrane protein	219	4476	1	2	3	3	3	3	4	3	5	3	1	3	3
NMC0006	putative glycerate dehydrogenase	954	4816	1	2	2	2	2	2	2	2	2	2	1	2	2
NMC0007 metG	methionyl-tRNA synthetase	2058	5843	1	2	2	2	2	2	2	2	2	2	1	3	3
NMC0008 glmS	glucosamine-fructose-6-phosphate aminotransferase [isomerizing]	1839	8016	1	2	2	2	2	2	2	2	2	2	1	2	2
NMC0009	putative lipoprotein	519	10290	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0010 gna33	outer membrane lipoprotein Gna33	1326	11226	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0011	putative integral membrane protein	840	12763	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0012	putative lipoprotein	1167	13599	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0013	possible membrane protein	1266	15029	1	2	1	1	3	3	1	1	1	1	1	1	1
NMC0014 phnA	putative phosphonoacetate hydrolase	330	16366	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0015 glmU	bifunctional GImU protein [includes: UDP-N-acetylglucosamine pyrophosphorylase (EC 2.7.7.23) (N-acetylglucosamine-1-phosphate uridylyltransferase); glucosamine-1-phosphate N-acetyltransferase (EC 2.3.1.57)]	1371	16772	1	1	1	1	1	1	2	1	1	1	1	1	1
NMC0016	conserved hypothetical protein	978	18197	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0017 tbpA	putative solute-binding periplasmic protein	1002	19233	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0018	putative inner membrane protein	849	20328	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0019	conserved hypothetical protein	486	21203	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0020 pilB	peptide methionine sulfoxide reductase	1569	21790	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0021 pilA	probable signal recognition particle protein	1266	23503	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0022	putative transposase	957	25628	1	2	2	1	1	1	1	2	2	2	1	2	2
NMC0023	conserved hypothetical protein (putative ATP-binding protein)	1188	26799	1	1	2	1	3	3	3	2	2	2	1	3	3
NMC0024	putative inner membrane protein	285	27987	1	2	2	3	3	3	3	2	2	2	1	2	2
NMC0025	TspB protein	1560	28574	1	1	1	X	1	1	1	1	1	1	1	2	3
NMC0027	putative inner membrane protein	303	30552	1	2	2	3	2	2	1	2	2	2	1	4	4
NMC0028	putative periplasmic protein	231	30882	1	1	1	2	1	1	1	1	1	1	1	1	1
NMC0029	conserved hypothetical protein	201	31339	1	1	1	1	1	1	1	1	1	1	1	1	1
NMC0030	conserved hypothetical protein	312	31543	1	2	3	1	1	1	1	3	3	3	1	4	2
NMC0031	conserved hypothetical protein	1000	31875	1	2	2	1	2	2	1	2	2	2	1	1	5

Distance matrix shows number of genes that are different between every pair of isolates

```
#NEXUS
[Distance matrix calculated by BIGSdb Genome Comparator (Thu Sep 12 09:43:44 2013)]
[Jolley & Maiden 2010 BMC Bioinformatics 11:595]
[Truncated loci excluded from analysis]
[Paralogous loci included in analysis]

BEGIN taxa;
  DIMENSIONS ntax = 13;

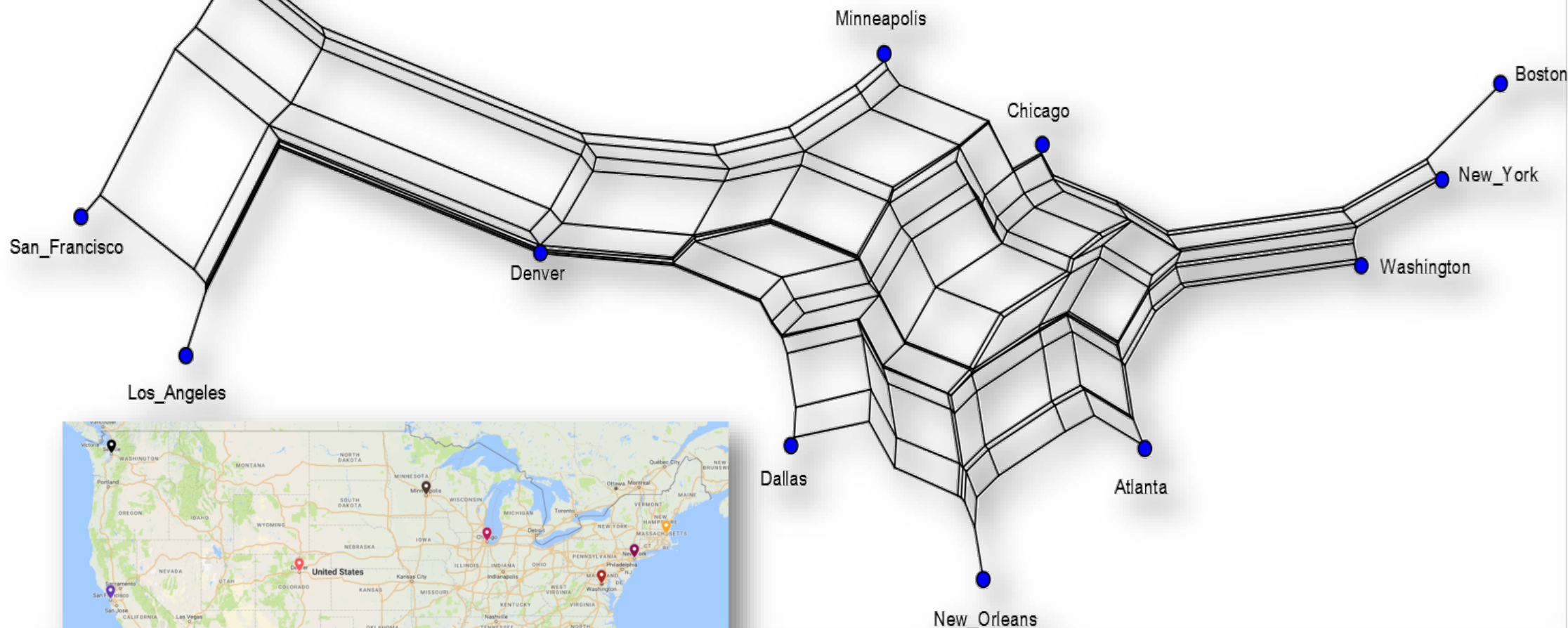
END;

BEGIN distances;
  DIMENSIONS ntax = 13;
  FORMAT
    triangle=LOWER
    diagonal
    labels
    missing=?
  ;
MATRIX
ref      0
644|L93/4286    626    0
662|2837      651    146    0
663|2839      603    189    221    0
664|2838      604    189    221    14    0
665|2845      366    671    689    645    645    0
666|2843      374    656    673    634    635    446    0
667|2842      379    662    675    639    639    441    28    0
669|2846      257    650    682    634    633    408    421    425    0
670|2840      657    153    33    235    235    696    677    683    692    0
671|2844      650    143    20    227    227    691    672    677    683    31    0
672|2847      649    147    21    223    221    690    670    676    683    34    26    0
698|FAM18      0      626    651    603    604    366    374    379    257    657    650    649    0
;
END;
```

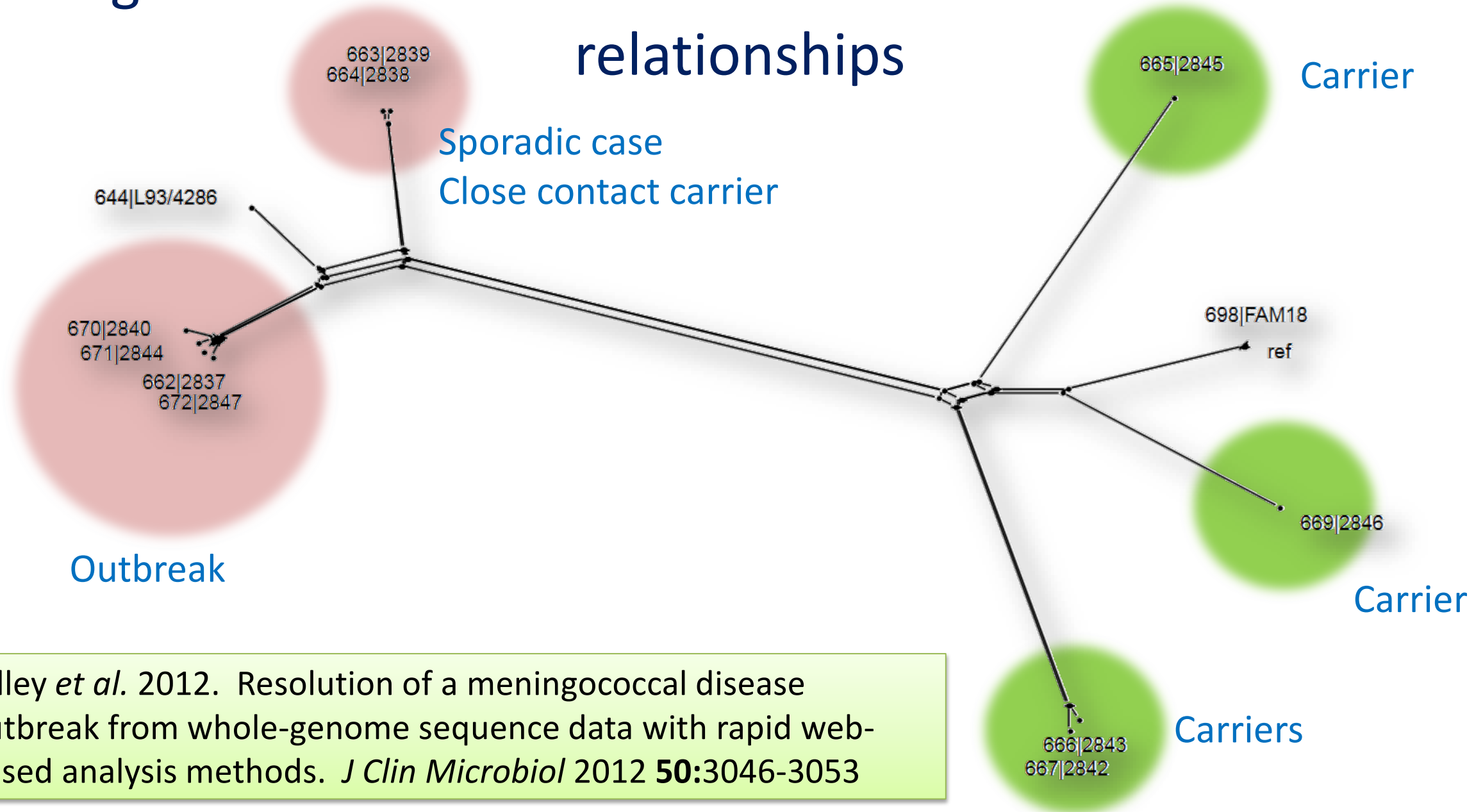
Genetic distances can be treated in the same way as geographic distances

New York	0												
Washington	204	0											
Boston	190	394	0										
Dallas	1372	1183	1551	0									
New Orleans	1170	966	1360	443	0								
Los Angeles	2448	2297	2594	1239	1671	0							
San Francisco	2569	2438	2696	1483	1924	348	0						
Atlanta	747	543	937	720	425	1935	2138	0					
Chicago	712	594	850	806	836	1744	1857	589	0				
Denver	1629	1491	1767	663	1082	831	948	1211	919	0			
Minneapolis	1017	932	1122	865	1054	1523	1583	909	355	699	0		
Seattle	2405	2324	2488	1682	2101	961	680	2181	1735	1021	1393	0	

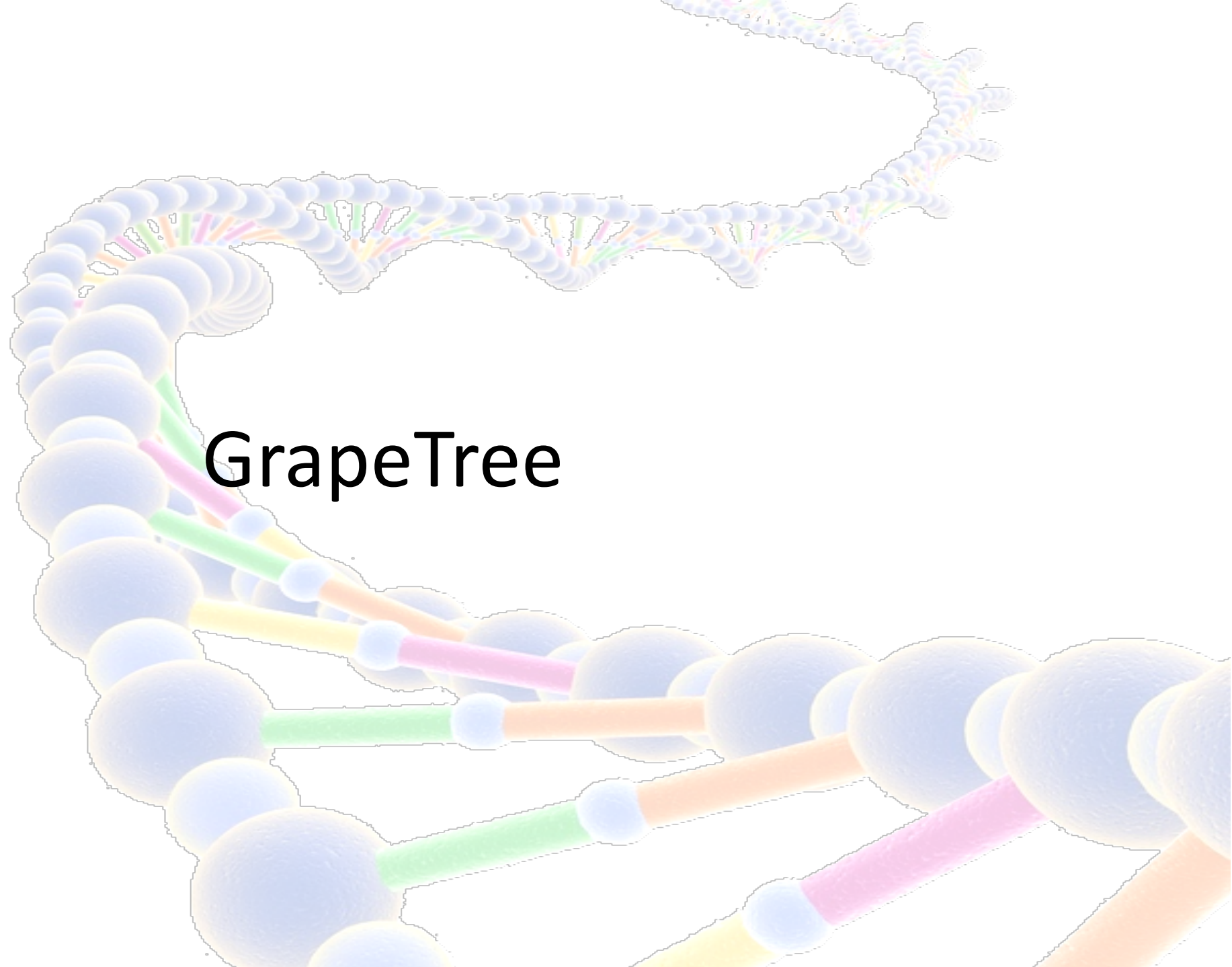
NeighborNet will visualise distances from a distance matrix



NeighborNet visualises distance matrix to resolve strain relationships



Jolley *et al.* 2012. Resolution of a meningococcal disease outbreak from whole-genome sequence data with rapid web-based analysis methods. *J Clin Microbiol* 2012 **50**:3046-3053



GrapeTree

GrapeTree analysis can be used to investigate phenotype/provenance compared to genotype

